

Deepfake & Synthetic Media Detection for Government Communications

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Live demo: https://hadiarahmani.github.io/GovVoiceShield_GISEC

Presentation: <https://sl1nk.com/fbkyc0m>

I. PROJECT OBJECTIVE AND RESEARCH CONTEXT

A. Problem, User, and Operational Boundary

The competition challenge is *Deepfake and Synthetic Media Detection for Government Communications*. We select audio to keep implementation and evaluation focused. The intended user is a communications officer or security reviewer receiving a purported official voice message. The decision supported is whether a recording warrants closer inspection and verification through a known official channel. A familiar voice is insufficient evidence that a message is authorized.

The proposed organizational adopter is a government communications or security team. The tool adds a structured screening step to manual listening; it does not replace an official verification channel. No procurement interest, analyst time saving, or organizational deployment benefit has been measured. The interface is intended to make model evidence and its limitations visible before a reviewer acts on a message.

Synthetic-speech detection is distinct from identifying a speaker, checking a statement's truth, detecting every edit, or proving legal authenticity. A genuine recording can contain false information, and generated speech can convey true information. The prototype therefore avoids an "authenticated" outcome and does not trigger automatic accusations or consequential actions.

B. Project Goals and Evaluation Questions

The project provides a complete local screening path for short audio and makes its result reviewable. A user can listen before reveal, compare a personal judgment with the model output, revisit suggested intervals, and distinguish a prediction from a documented source label. Dataset identifiers, split assignments, integrity hashes, parameters, and evaluation outputs preserve reproducible evidence.

The evaluation asks whether the browser reproduces the Python model; how the frozen classifier performs on controlled, unused test recordings; how noise and MP3 compression change its results; and whether the interface handles invalid or unusual input. Accuracy, precision, recall, ROC-AUC, confusion matrices, calibration error, abstention coverage, and software checks provide complementary evidence. No single score establishes operational readiness.

C. Threat and Decision Model

The motivating threat is audio that appears to come from an official source but may have been generated, converted, edited, compressed, or replayed. Its sender, synthesis engine, recording channel, and original file may be unknown. Missed synthetic speech can encourage trust in fraud, while a false alarm can undermine a genuine communication.

GovVoice Shield therefore acts as a triage aid. The reviewer remains responsible, and consequential messages should be checked through a known official channel regardless of the model score. The study assumes that the browser executes the supplied code and model unchanged; it does not defend against a compromised device, replaced model, or input designed specifically against the classifier.

D. Research Foundation and Contribution

Established anti-spoofing research motivates acoustic classification and neural alternatives [1], [2]. Dataset shift remains central. Müller et al. found that raw-waveform detectors achieving as little as 1.2% equal-error rate (EER) on ASVspoof reached 33.9% EER on their in-the-wild data [3]. Speaker-to-microphone replay creates an additional challenge [4].

The present contribution is the integration of local processing, a blind listening workflow, separately calibrated scoring, and honest source-label reveal. It is not a claim of a novel classifier architecture or superior benchmark performance. The research question is whether a compact, inspectable detector can provide useful screening evidence while communicating the boundaries of its measured results.

II. PROPOSED SOLUTION AND ARCHITECTURE

A. Human-in-the-Loop Workflow

The user selects a demo clip, uploads WAV or MP3 audio, or records a short microphone sample. Playback and a human/synthetic/not-sure guess precede the model reveal. Demo source labels remain separate from predictions, including when the model is wrong. Uploaded and microphone recordings have unknown labels unless their provenance is established independently.

The interface offers likely human, likely synthetic, or uncertain outcomes; a study-calibrated estimate; review timestamps; and a downloadable JSON assessment. Eight identifier-selected benchmark clips and two fictional stock-voice announcements form the ten-sample reel. A separate human false-alarm example demonstrates a known error and is not counted as an independent test.

B. Current Detection Approaches vs. Our Approach

Most published anti-spoofing systems are deep, end-to-end classifiers such as AASIST and wav2vec-based detectors [1], [2], run server-side and reported as a single opaque score that can collapse on unseen, in-the-wild audio [3]. GovVoice Shield instead runs a compact, engineered-feature SVM entirely in the browser, exposing signal evidence, calibrated confidence, and abstention alongside a blind human guess before reveal — prioritizing an inspectable, reviewer-led decision over raw benchmark accuracy.

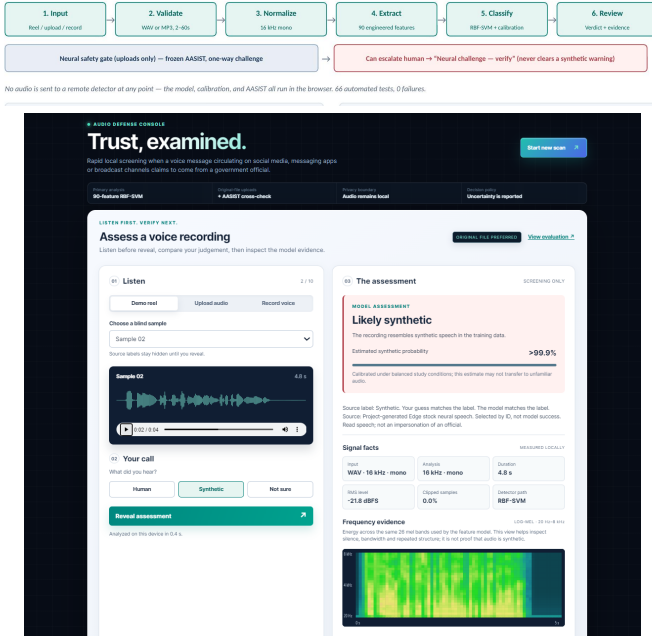


Fig. 1. Implemented system workflow (top) and the current reveal view for a demo-reel sample assessed as likely synthetic (bottom). The interface reports signal facts, log-mel frequency evidence, and exploratory review intervals without presenting them as forensic proof.

The local path validates an upload or 3–15 s capture, decodes 16-kHz mono audio, extracts and scales 90 features, applies the RBF SVM and calibration gates, and presents the result for review. Two-second windows provide exploratory listening hints. No audio is sent to a remote detector.

C. Acoustic Representation

The waveform is converted to 16-kHz mono. Mean subtraction and peak normalization precede feature extraction. We use 400-sample (25-ms) Hamming frames, a 160-sample (10-ms) hop, a 512-point FFT, 26 mel filters, and 13 mel-frequency cepstral coefficients (MFCCs). For each coefficient, means and

standard deviations are computed for the original sequence and its first and second temporal differences. This produces $13 \times 2 \times 3 = 78$ values. Means and standard deviations of zero-crossing rate, spectral centroid, bandwidth, roll-off, flatness, and frame RMS add 12 values, giving 90 features.

The standardization parameters are fitted only on training features. The SVM margin for a standardized feature vector z is

$$f(z) = b + \sum_{i \in \mathcal{S}} \alpha_i \exp(-\gamma \|z - z_i\|_2^2), \quad (1)$$

where $\mathcal{S}$ denotes support vectors and the signed coefficients incorporate class labels. The browser evaluates the exported parameters rather than calling a model API. Python and JavaScript feature/prediction parity is tested numerically.

D. Calibration, Abstention, and Evidence

A logistic mapping $\hat{p} = \sigma(af(z) + c)$ is fitted on a separate calibration partition, following the sigmoid approach introduced for SVM outputs by Platt [5]. It estimates synthetic class membership under the study’s balanced domain/class weighting. It is not an operational probability of authenticity. Conservative calibration-selected thresholds permit human or synthetic decisions; intermediate values return uncertain. Unsupported threshold regions are disabled. Observed calibration precision does not guarantee future precision.

Quality rules reject silence or invalid values and abstain for heavy clipping, very short recordings, or extreme standardized feature distance. These heuristics do not recognize every distribution shift. A separate two-second-window analysis with a one-second hop displays raw margins to guide another listen. Whole-recording feature statistics and short-window statistics differ; window scores are neither a vote defining the final verdict nor calibrated localization evidence.

The reveal view also reports the original format, sample rate and channels; the fixed 16-kHz mono analysis rate; duration; RMS level; clipping; and a 26-band log-mel energy map. The map uses the same mel-band definition as the feature model, but it is a visualization of signal energy rather than a causal model explanation or authenticity proof.

E. Implementation and Focus

The deployed model uses engineered features and an SVM trained for this project. User uploads also receive a frozen pretrained AASIST safety check [1]. It may challenge a human-leaning primary result, but it cannot clear an RBF-SVM synthetic warning. Its weights were not fine-tuned by the team.

ENF matching, voice identity, blockchain, video analysis, and general edit attribution are outside the current implementation. Electrical hum, pitch, loudness, or mono conversion alone do not establish synthesis. Video analysis is reserved for a separately validated future module.

III. SOLUTION VALIDATION

A. Data Sources and Recorded Provenance

The second-version experiment comprises 2,106 recordings: 1,866 clips from the public six-platform collection [6], 120 human LibriSpeech clips [7], and 120 project-generated, text-matched stock-voice counterparts. Public synthesis sources include Polly, ElevenLabs, Kokoro, Hume, Luvvoice, and Speechify. The collection describes December 2024 material; it is not a comprehensive sample of current attacks. Repository revisions, identifiers, source groups, file hashes, label origins, and splits are recorded.

LibriSpeech labels follow the human-read corpus. Generated labels follow the recorded generation procedure, rather than another detector’s prediction. Stock Microsoft voices were used without cloning an official. The two government-style announcements are explicitly fictional. Public corpus labels and licence declarations are source-provided, not independent forensic or rights audits. LibriSpeech and the contemporary collection declare CC-BY-4.0; generated output remains subject to its provider’s terms. Archived datasets/components retain their separate notices.

TABLE I
FROZEN V2 PARTITION COUNTS BEFORE AUGMENTATION

Partition	Human	Synthetic	Total
Training	544	549	1,093
Model selection	115	132	247
Calibration	142	139	281
Final test	252	233	485
Total	1,053	1,053	2,106

B. Separation and Leakage Controls

Training, model selection, calibration, and test assignments were fixed before final-test feature access. Related public source recordings are grouped. Public synthetic numeric identifiers are conservatively grouped across platforms where applicable, and Hume is completely excluded from development. However, unknown public speaker identities and potentially shared scripts prevent a claim of complete speaker/text independence for that diagnostic.

The paired LibriSpeech/stock-voice study separates human speakers, text identifiers, and synthetic stock voices across development and test. Exact normalized text overlap and cross-split exact-file duplicates were checked. The same Edge synthesis engine nevertheless appears in training and test. Thus, unseen stock voices are not equivalent to an unseen generation system. Neither group separation nor a hash check rules out every possible correlation or corpus artifact.

Training includes clean and deterministic 20-dB white-noise variants; variants remain with their source partition. Equal weighting of domain/class cells reduces domination by the larger public collection. Scaling is fitted on training data only. Nine RBF SVM configurations, $C \in \{1, 10, 100\}$ and $\gamma \in \{0.003, 0.01, 0.03\}$, are compared using the mean tuning

ROC-AUC across paired and public domains. Calibration is subsequently fitted on its separate partition [8].

C. Locked Test and Endpoints

The primary endpoint uses 80 paired recordings: 40 human and 40 synthetic. The remaining 405 public recordings form a separate diagnostic, not a pooled headline metric. After model and threshold hashes were frozen, evaluation computed accuracy, synthetic precision/recall, F1, ROC-AUC, Brier score, confusion matrices, and abstention coverage. Fixed stress conditions add 20-dB white noise or encode/decode 32-kbps MP3. These tests do not reproduce every phone, room, social-media codec, or microphone channel.

Tables report forced binary decisions at $\hat{p} \geq 0.5$. Interface performance is reported separately because an uncertain outcome is not a correct binary prediction. Coverage is the fraction for which the interface makes a decision; accepted-only accuracy excludes abstentions. The clean primary Wilson interval is descriptive because shared voices and paired scripts violate independent-clip assumptions.

D. Software, Privacy, and Error Handling

Audio upload is limited to supported formats, 2–60 seconds, and 20 MB. Parsing checks precede expensive analysis; decoded samples are checked for finite values, duration, and silence. Analysis runs in a worker with a 45-second timeout. Input changes invalidate stale work, and object URLs are released. User-controlled text is rendered as text rather than inserted HTML. A restrictive content policy bounds resource origins.

Microphone access follows an explicit Start action. Capture stops at 15 seconds and releases tracks on stop or cancellation; leaving the tab cancels active capture. Browser permission and codec support remain device-dependent. Captured audio is converted locally to PCM for the existing pipeline. No microphone recording is automatically assigned a human label.

The core, model, recorder, source-policy, and neural suites contain 17, 13, 10, 11, and 15 passing checks, respectively: 66 checks with no failures. These include numerical parity, bounded frequency-map output, WAV/MP3 metadata inspection, ten-file worker inference, malformed input, repeatability, permission cancellation with mocked devices, and PCM round-tripping. They are not a penetration test or complete browser acceptance test. A repeated 60-second waveform remained below the interface timeout in local testing; this is not a cross-device latency guarantee.

IV. RESULTS AND OBSERVED LIMITATIONS

A. Controlled Primary Results

Table II reports the frozen v2 primary test. Clean accuracy is 97.5%, with synthetic recall of 100% and precision of 95.24%. Its F1 is 0.976 and Brier score is 0.017. The confusion matrix $[[38, 2], [0, 40]]$ shows that the two clean errors are human recordings classified as synthetic. The descriptive clip-level accuracy interval is approximately 91.34–99.31%; it should not be read as a deployment guarantee.

TABLE II
PRIMARY TEST: 80 CLIPS; SYNTHETIC IS THE POSITIVE CLASS

Condition	Acc. (%)	Prec. (%)	Rec. (%)	AUC
Clean	97.50	95.24	100.00	0.999
Noise, 20 dB	93.75	90.70	97.50	0.994
MP3, 32 kbps	92.50	88.64	97.50	0.979

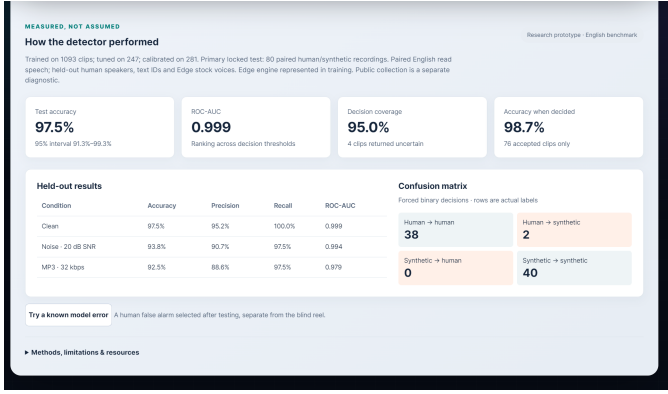


Fig. 2. Prototype evaluation view showing the frozen primary metrics, abstention coverage, stress-test results, and confusion matrix.

At the interface thresholds and quality gates, 76 of 80 primary clips receive a decision: 95% coverage, with four uncertain outcomes. Accepted-only accuracy is 75/76, or 98.68%. Reporting that value without coverage would obscure abstentions and overstate overall performance. Under MP3 stress, five human clips are flagged synthetic and one synthetic clip is missed. These errors demonstrate that channel changes can alter both false-alarm and miss rates.

B. Separate Public-Audio Diagnostic

The public diagnostic contains 212 human and 193 synthetic recordings. Clean accuracy is 94.81%, precision 94.79%, recall 94.30%, and ROC-AUC 0.993. Its confusion matrix is $[[202, 10], [11, 182]]$. Added noise reduces accuracy to 92.10%; MP3 reduces it to 86.42%. In the MP3 condition, 52 human recordings become false alarms. This deterioration is important even when ranking metrics remain comparatively high.

Hume, excluded from model development, contributes 116 synthetic test clips, of which 106 are detected at the forced-binary threshold (91.38% recall). This is a single-class, single-platform diagnostic, not balanced accuracy or general unseen-

generator performance. Unverified public speaker/script relationships and limited source groups further constrain interpretation.

C. Post-Test Failure Investigation

After the experiment, user testing reported synthetic uploads and speaker-to-microphone recordings being labelled human. One supplied 12.771-second MP3 was decoded and scored locally, reproducing an RBF-SVM synthetic estimate of approximately 0.027%. A frozen AASIST check produced 99.78% on the same decoded audio, so the upload policy exposes both values and challenges the human clearance. Applied post-test to the preserved 80-clip primary set, this asymmetric gate accepted 75 clips and was correct on 74; AASIST alone achieved 56.25% accuracy and 0.506 ROC-AUC. These exploratory figures follow failure inspection, do not make the supplied label an independent benchmark, and do not replace the frozen v2 results.

In that local decode, ten of eleven short-window margins were positive while the whole-file verdict was human. This is not a mathematical contradiction: the SVM receives different whole-file and window feature statistics, and its final decision does not aggregate window votes. It nevertheless creates a communication problem if users treat the timeline as an independent, validated detector.

The sample had no measured clipping, RMS approximately 0.139, and standardized feature distance 1.227 against an outlier cutoff of 11.190. Existing quality/outlier rules therefore did not flag it. These measurements identify a limitation of those rules; they do not establish the precise acoustic cause or verify that the recording is synthetic. Pitch, loudness, or another detector's verdict cannot provide that missing provenance.

D. Scope of the Claims

The implemented software processes actual waveforms; it does not choose outputs from filenames or demo labels. Its functioning interface and passing tests do not establish dependable real-world classification. The observed study accuracy applies to the documented subsets. It does not validate Arabic, arbitrary new engines, music, mixed human/AI tracks, phone channels, or official-speaker authentication. No dedicated speech-activity detector is implemented, so non-speech input may still receive a score.

The subsequent failures make replacement-model development exploratory. Existing test results must remain frozen and labelled as v2. A new model, aggregation rule, or uncertainty gate requires separate development data and a fresh locked evaluation; correcting inspected files alone is not evidence of generalization.

V. DISCUSSION, FUTURE WORK, AND CONCLUSION

A. Results in Context

The controlled results show that a compact detector can support an interactive screening demonstration, but they also show why benchmark accuracy cannot be treated as deployment accuracy. Compression changes both miss and false-alarm rates in this study. Published evidence reaches the same broader conclusion: detectors with low benchmark EER can deteriorate sharply on in-the-wild recordings [3]. In replay experiments, W2V2-AASIST increased from 4.7% EER on original audio to 18.2% after playback and re-recording; room-impulse-response augmentation reduced the replay EER to 11.0%, but did not restore the original result [4]. These are published results, not measurements of GovVoice Shield, and they motivate rather than validate the next model.

The local listen-guess-reveal workflow remains useful for research demonstrations because it separates the participant's judgment, the model assessment, and the known source label. The known false-alarm example and the reported post-test failures should remain visible. This makes the prototype inspectable while avoiding an unsupported claim that a confidence value authenticates a speaker or message.

B. Demonstration and Reproducibility

A demonstration should select a blind reel sample, play it, collect a guess, reveal the assessment and source label, and replay an indicated interval. The separate known human false alarm should also be shown. A supported upload or microphone sample and a recoverable input error demonstrate interaction and repeat operation. The presenter should explain what was trained, how labels were obtained, and why the confidence estimate may fail on unfamiliar recordings.

The source package contains the application, exported model, manifests, locks, predictions, test results, and run instructions. The GitHub Pages demonstration is available at https://hadiarahmani.github.io/GovVoiceShield_GISEC, and the source repository is https://github.com/HadiaRahmani/GovVoiceShield_GISEC. A local static server can be started from the project folder with `python -m http.server 8765 -bind 127.0.0.1 -directory dist`. The browser then opens `http://127.0.0.1:8765`. HTTPS or localhost is required for microphone access. Local inference needs no detector backend or remote model service.

C. Future Work

The next audio phase will preserve the v2 evidence and confirm or reject the asymmetric safety gate on a newly frozen test. It will collect independently labelled genuine and generated recordings with generator version, consent or licence records, original files, compressed derivatives, and physical recordings. Related versions will remain in the same partition. ReplayDF can support replay diagnostics, subject to its CC-BY-NC-4.0 non-commercial licence and verification of its source metadata and partition independence [4]. Model choice, calibration, and abstention rules must be frozen before a new

final test. Government-domain and Arabic recordings should be evaluated before corresponding claims are made.

A later video module could examine spatial and temporal evidence such as face-landmark consistency, lip-audio synchronization, frame-level artifacts, and compression robustness. Its dataset, identity permissions, training split, calibration, and locked evaluation must be independent of the audio study. Audio and video scores should only be fused after each modality is validated separately; disagreement should produce an uncertain outcome for human review. This staged design extends the system to synthetic media without presenting video detection as an implemented capability.

D. Conclusion

GovVoice Shield is a functioning, inspectable research prototype with a documented local workflow and reproducible subset measurements. The frozen v2 achieved strong controlled-test accuracy, but compression sensitivity and reported post-test failures prevent a claim of dependable unfamiliar-audio detection. The immediate priority is representative, independently labelled evaluation and improved confidence communication, not additional interface complexity. A successful research demonstration must communicate both the measured progress and the unresolved reliability boundary.

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